THE COMPARISON MAP BETA BETWEEN THE STALK OF K AND THE COHOMOLOGY OF THE MILNOR FIBRE IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to the strictly henselian case with the observation that local acyclicity is preserved, the description of the relevant fibre product, the identification of the stalk map with alpha, and the concluding stalk criterion.

Editorial note. The label lemma-beta-is-isomorphism names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to maps constructed earlier in the chapter and is delivered verbatim; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Local acyclicity of f relative to K via the maps alpha and beta. Same chapter, the definition of local acyclicity using the maps alpha_{K,x,t} and beta_{K,s,t} and their relation. Supplies the map beta whose isomorphism is asserted and the description of alpha that it is compared with.
- Stalks and pullback: i^{-1} preserves stalks. Same chapter, the lemma identifying the stalk of a pullback with the stalk of the original sheaf. Supplies the first equality in the computation of the stalk of i^{-1}beta.
- Cohomology of the Milnor fibre via higher direct images. Same chapter, the theorem describing the stalk of Rh_* as the cohomology of the fibre F_{x,t}. Supplies the identification of Rh_*(K|_{X_t})_{x} with R Gamma(F_{x,t}, K).
- A map of sheaves is an isomorphism iff it is on stalks. Same chapter, the exactness-on-stalks theorem for the etale topology. Supplies the final reduction from sheaves to stalks.

Source reference. Original reference 0GJP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19110–19110. Prerequisite pointer invoked at proof

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line(s) 19459; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GJS: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19289–19289. Prerequisite pointer invoked at proof line(s) 19431; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 008H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4649–4649. Prerequisite pointer invoked at proof line(s) 19444; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Prerequisite pointer invoked at proof line(s) 19461; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Prerequisite pointer invoked at proof line(s) 19456; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The map $i^{-1}\beta_{K,\overline{s},\overline{t}}$ is an isomorphism.*

Proof. The construction of the maps h, i, $\beta_{K,\overline{s},\overline{t}}$ only depends on the base change of X and K to $\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$. Thus we may and do assume that S is a strictly henselian scheme with closed point $\overline{s}$. Observe that the local acyclicity of f relative to K is preserved by this base change (for example by Lemma 0GJS or just directly by comparing strictly henselian rings in this very special case).

Let $\overline{x}$ be a geometric point of $X_{\overline{s}}$. Or equivalently, let $\overline{x}$ be a geometric point whose image by f is $\overline{s}$. Let us compute the stalk of $i^{-1}\beta_{K,\overline{s},\overline{t}}$ at $\overline{x}$. First, we have

$$(i^{-1}\beta_{K,\overline{s},\overline{t}})_{\overline{x}} = (\beta_{K,\overline{s},\overline{t}})_{\overline{x}}$$

since pullback preserves stalks, see Lemma 008H. Since we are in the situation $S = \mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$ we see that $h : X_{\overline{t}} \to X$ has the property that $X_{\overline{t}} \times_X \mathrm{Spec}(\mathcal{O}^{sh}_{X,\overline{x}}) = F_{\overline{x},\overline{t}}$. Thus we see that

$$(\beta_{K,\overline{s},\overline{t}})_{\overline{x}} : K_{\overline{x}} \longrightarrow Rh_*(K|_{X_{\overline{t}}})_{\overline{x}} = R\Gamma(F_{\overline{x},\overline{t}}, K)$$

where the equal sign is Theorem 03Q9. It follows that the map $(\beta_{K,\overline{s},\overline{t}})_{\overline{x}}$ is none other than the map $\alpha_{K,\overline{x},\overline{t}}$ used in Definition 0GJP. The result follows as we may check whether a map is an isomorphism in stalks by Theorem 03PU. □